

Definite integrals



1 Evaluate the following definite integrals:

a $\int_0^3 (4x + 5) dx$

c $\int_2^4 (6x + 5) dx$

e $\int_{-4}^5 (-8x + 3) dx$

g $\int_{-2}^0 9x^2 dx$

i $\int_{-2}^1 (x^2 + 4) dx$

k $\int_4^6 (9x^2 + 2x + 7) dx$

m $\int_0^4 5x^{\frac{3}{2}} dx$

o $\int_6^{11} \sqrt{x-2} dx$

q $\int_3^4 (2x + 3)^3 dx$

s $\int_1^2 \frac{x^5 - x^{-2}}{x^2} dx$

b $\int_{-2}^0 (10x + 4) dx$

d $\int_3^4 (-4x + 3) dx$

f $\int_{-10}^{-6} (-5x + 8) dx$

h $\int_{-1}^3 9x^2 dx$

j $\int_{-1}^2 (9x^2 + 1) dx$

l $\int_{-4}^2 x(x - 4) dx$

n $\int_{-3}^6 x(x + 3)(x - 6) dx$

p $\int_3^6 (\sqrt{x-2} + 5) dx$

r $\int_{-4}^2 \left((x + 2)^3 + 3 \right) dx$

Properties of definite integrals

2 Consider the function $f(x) = x^2$. Find the value of the following:

a $\int_1^1 f(x) dx.$

b $\int_5^5 f(x) dx.$

c $\int_{-2}^{-2} f(x) dx.$

d $\int_a^a f(x) dx.$

3 Consider the function $f(x) = 2x$.

a Find the value of $\int_0^3 f(x) dx.$

b Find the value of $\int_0^3 5f(x) dx.$

c Find the value of $5 \int_0^3 f(x) dx.$

d State the property of definite integrals demonstrated by parts (b) and (c).

4 Given that $\int_{-2}^7 f(x) dx = 2$, find $\int_{-2}^7 5f(x) dx.$

5 Given that $\int_{-1}^6 f(x) dx = 3$, find $\int_{-1}^6 (9f(x) - 2) dx.$

6 Consider the function $f(x) = 6x$.



- a Find the value of $\int_4^8 f(x) dx$.
- b Find the value of $\int_8^4 f(x) dx$.
- c State the property of definite integrals demonstrated by parts (a) and (b).

7 Given that $\int_4^6 f(x) dx = 3$, find the values of the following:

- a $\int_6^4 f(x) dx$
- b $\int_4^6 3f(x) dx$
- c $\int_4^6 (f(x) + x) dx$

8 Given that $\int_{-1}^2 f(x) dx = 4$ and $\int_2^8 f(x) dx = 8$, find the values of the following:

- a $\int_{-1}^8 f(x) dx$
- b $\int_8^{-1} f(x) dx$
- c $\int_{-1}^2 2f(x) dx + \int_2^8 3f(x) dx$

9 Given that $\int_{-1}^3 f(x) dx = 5$ and $\int_2^3 f(x) dx = 2$, find the values of the following:

- a $\int_{-1}^2 f(x) dx$
- b $\int_3^{-1} f(x) dx$
- c $2 \int_{-1}^2 f(x) dx + \int_2^3 3f(x) dx$

Signed areas

10 Consider the graph of $y = f(x)$.

- a Find the value of $\int_0^2 f(x) dx$.
- b Find the value of $\int_2^5 f(x) dx$.
- c Find the value of $\int_5^6 f(x) dx$.
- d Hence state the area bounded by the function and the x -axis.
- e Write a single definite integral to represent the area bounded by the function and the x -axis.



11 Consider the graph of $y = f(x)$.

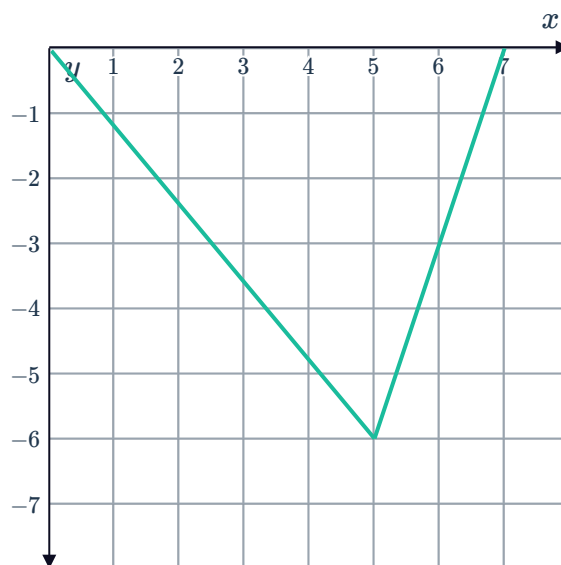


- a Find the value of $\int_0^4 f(x) dx$.
- b Find the value of $\int_4^6 f(x) dx$.
- c Find the value of $\int_6^8 f(x) dx$.
- d Hence state the area bounded by the function and the x -axis.
- e Write a single definite integral to represent the area bounded by the function and the x -axis.



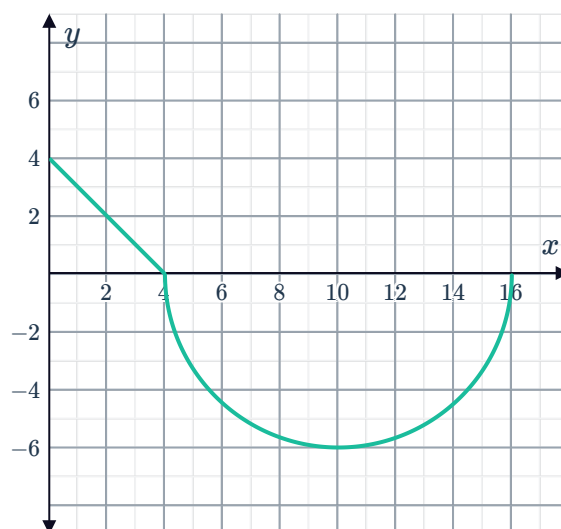
12 Consider the graph of $y = f(x)$.

- a Find the value of $\int_0^5 f(x) dx$.
- b Find the value of $\int_5^7 f(x) dx$.
- c Hence find the value of $\int_0^7 f(x) dx$.
- d State the area bounded by the function and the x -axis.
- e Write a single definite integral to represent the area bounded by the function and the x -axis.



13 Consider the graph of $y = f(x)$.

- a Find the value of $\int_0^4 f(x) dx$.
- b Find the exact value of $\int_4^{16} f(x) dx$.
- c Hence find the exact value of $\int_0^{16} f(x) dx$.
- d Calculate the exact area bounded by the curve, the x -axis and the y -axis.
- e Write an expression for the exact area from part (d) as a sum or difference of definite integrals.



14 Consider the function $y = -5$.



- a State whether the graph is above or below the x -axis.
- b Calculate $\int_{-4}^3 (-5) dx$.
- c Hence find the area bounded by the curve, the x -axis and the bounds $x = -4$ and $x = 3$.

15 Consider the function $y = 2x - 8$.

- a Find the x -intercept of the function.
- b State the values of x for which the graph is above the x -axis.
- c Calculate $\int_{-2}^4 (2x - 8) dx$.
- d Hence find the area bounded by the line, the x -axis and the bounds $x = -2$ and $x = 4$.

16 Consider the function $y = (x - 3)(x - 9)$.

- a Find the x -intercepts of the function.
- b State the values of x for which the graph is below the x -axis.
- c Calculate $\int_3^9 (x - 3)(x - 9) dx$.
- d Hence find the area bounded by the curve, the x -axis and the bounds $x = 3$ and $x = 9$.

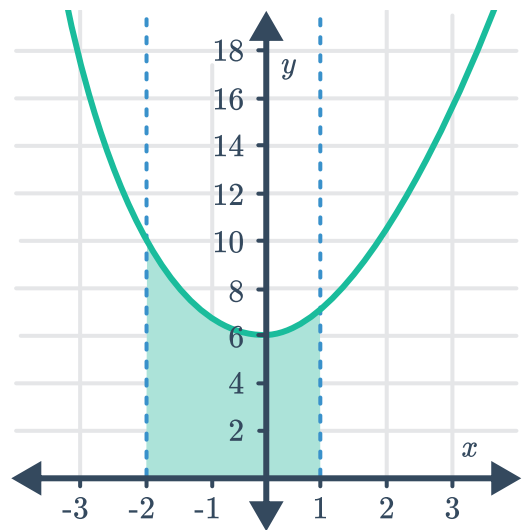
17 Consider the function $y = -(x + 2)(x + 8)$.

- a Find the x -intercepts of the function.
- b State the values of x for which the graph is above the x -axis.
- c Calculate $\int_{-8}^{-2} -(x + 2)(x + 8) dx$.
- d Hence find the area bounded by the curve, the x -axis and the bounds $x = -8$ and $x = -2$.

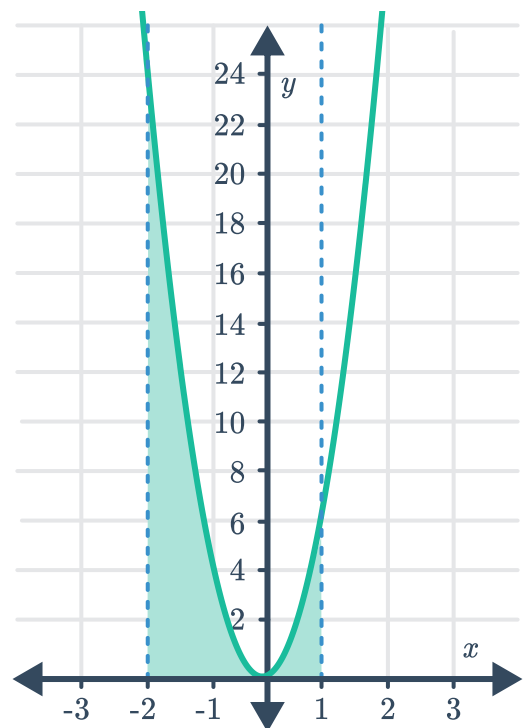


Area under curve

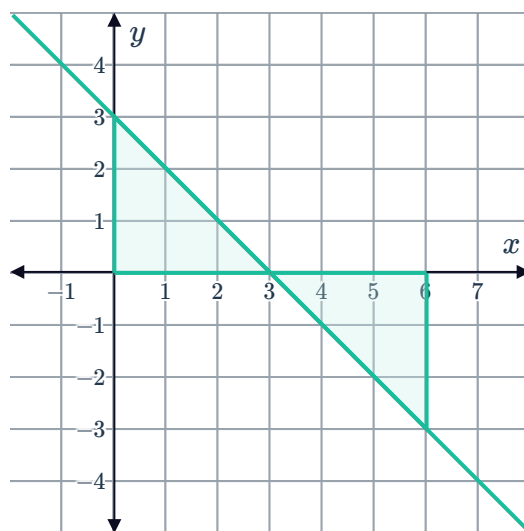
- 18 Consider the graph of the curve $y = x^2 + 6$.
Find the exact area of the shaded region.



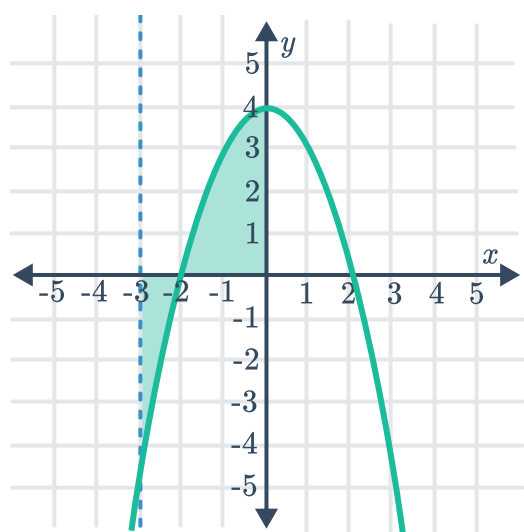
- 19 Consider the graph of the curve $y = 6x^2$.
Find the exact area of the shaded region.



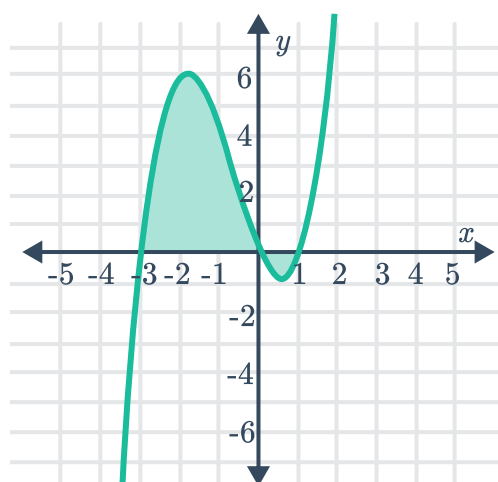
- 20** Consider the graph of the line $x + y = 3$.
Find the exact area of the shaded region.



- 21** Consider the graph of the curve $y = 4 - x^2$.
Find the exact area of the shaded region.



- 22** Consider the graph of the curve
 $y = x(x - 1)(x + 3)$.
Find the exact area of the shaded region.



23 Consider the given functions.



- i Sketch a graph of the function.
- ii Hence calculate the exact area bounded by the curve, $x = 1$, $x = 3$ and the x -axis.

a $y = 2x + 3$ **b** $y = -2x + 8$ **c** $y = 4x - x^2 - 3$

24 Consider the function $y = (x - 1)^2 (x + 3)$.

- a** Sketch a graph of the function.
- b** Hence calculate the exact area bounded by the curve, x -axis and the lines $x = -1$ and $x = 2$.

25 Consider the function $y = (x + 1)^3 + 2(x + 1)^3 + 2$.

- a** Sketch a graph of the function.
- b** Hence calculate the exact area bounded by the curve, x -axis, and the lines $x = -2$ and $x = 1$.

26 Consider the function $y = \sqrt{x + 1}$.

- a** Sketch a graph of the function.
- b** Hence calculate the exact area bounded by the curve, the x -axis, and the line $x = 3$.

27 For each of the following functions:

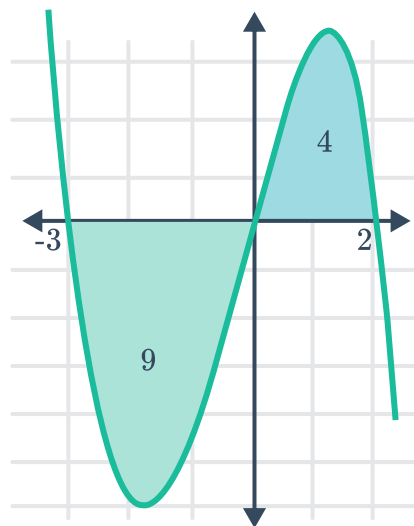
- i Sketch a graph of the function.
- ii Hence determine the exact area bounded by the curve and the x -axis.

a $y = x(x - 1)(x + 3)$

b $y = (x - 1)(x + 2)(x + 3)$

c $y = x(x - 1)$

- 28** Consider the function $f(x)$ as shown. The numbers in the shaded regions indicate the area of the region.
Find the value of the following:

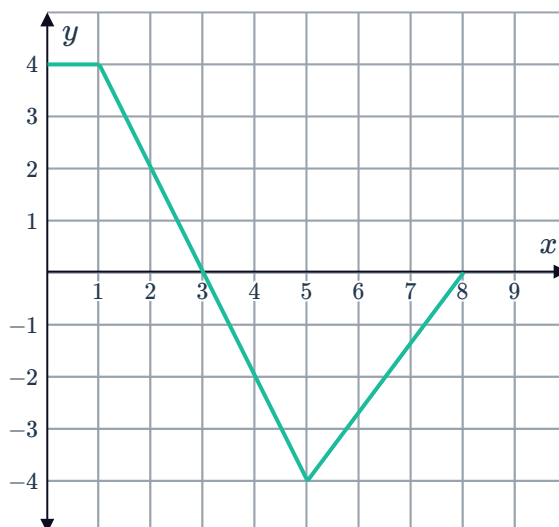


- a $\int_{-3}^0 f(x) dx$
- b $\int_{-3}^2 f(x) dx$
- c The area enclosed by the curve and the x -axis for $x < 0$.
- d The area enclosed by the curve and the x -axis.

Area as an absolute value

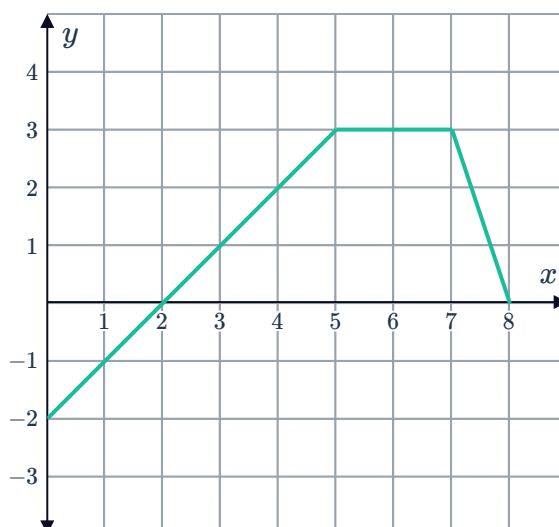
- 29** Consider the graph of $y = f(x)$.

- a Find the value of $\int_0^3 f(x) dx$.
- b Find the value of $\int_3^8 f(x) dx$.
- c Hence calculate $\int_0^8 f(x) dx$.
- d Calculate the area bounded by the function, the x -axis and the y -axis.
- e Write an expression for the area from part (d) as a sum or difference of definite integrals.
- f Write an expression for the area from part (d) using an absolute value sign.

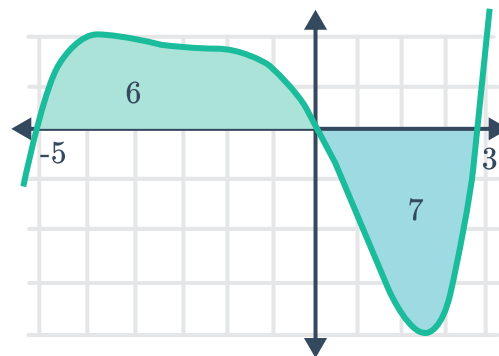


- 30** Consider the graph of $y = f(x)$.

- a Find the value of $\int_0^8 f(x) dx$.
- b Calculate the area bounded by the function, the x -axis and the y -axis.
- c Write an expression for the area from part (b) as a sum or difference of definite integrals.
- d Write an expression for the area from part (b) using an absolute value sign.

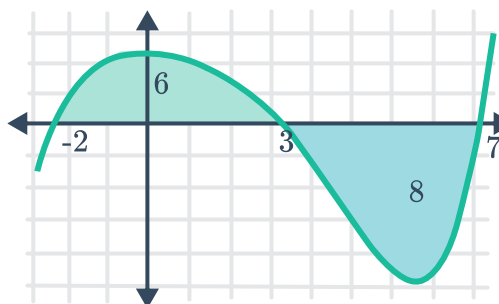


- 31** Consider the function $f(x)$ as shown. The numbers in the shaded regions indicate the area of the region. Find the following:



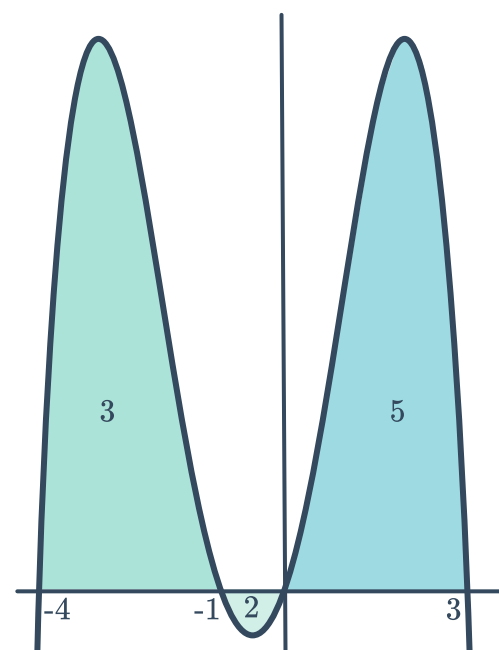
- a $\int_{-5}^3 f(x) dx$
- b $\left| \int_{-5}^3 f(x) dx \right|$
- c $\int_{-5}^3 |f(x)| dx$
- d The area enclosed by the curve and the x -axis.

- 32** Consider the function $f(x)$ as shown. The numbers in the shaded regions indicate the area of the region. Find the following:



- a $\int_{-2}^7 f(x) dx$
- b $\int_3^7 (-f(x)) dx$
- c $\int_{-2}^7 2f(x) dx$
- d $\int_3^{-2} f(x) dx + \int_3^7 f(x) dx$
- e $\int_{-2}^7 |f(x)| dx$
- f The area enclosed by the curve and the x -axis.

- 33** Consider the function $f(x)$ as shown. The numbers in the shaded regions indicate the area of the region. Find the following:

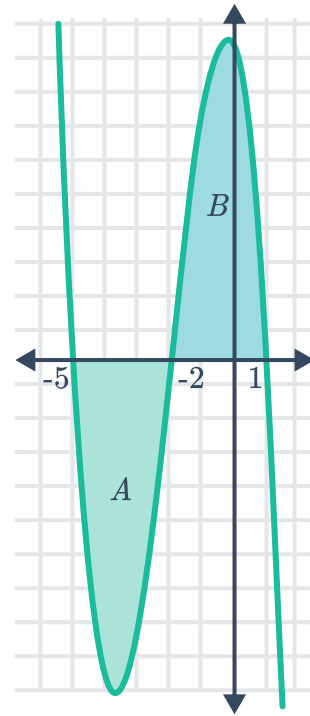


- a $\int_{-4}^0 f(x) dx$
- b $\left| \int_{-4}^3 f(x) dx \right|$
- c $\int_{-1}^0 2f(x) dx + \int_3^0 f(x) dx$
- d The area enclosed by the curve and the x -axis.
- e $\int_{-4}^3 |f(x)| dx$
- f $\int_{-4}^3 (f(x) + x^2) dx$ given that the definite integral $\int_{-4}^3 x^2 dx = \frac{91}{3}$.



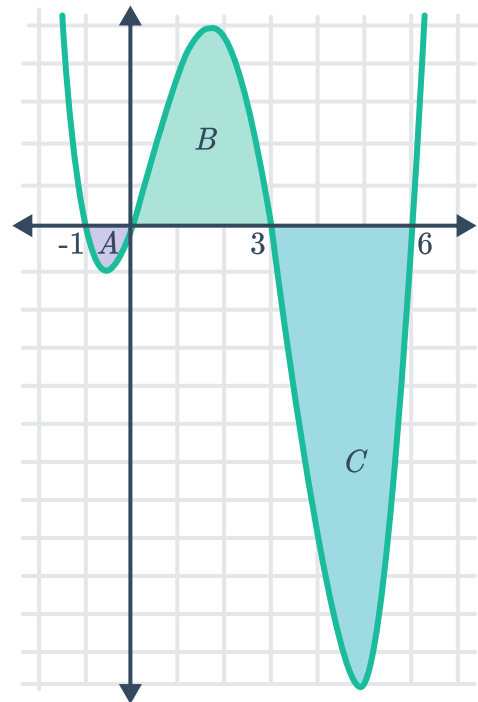
- 34** Consider the function $f(x)$ as shown. The numbers in the shaded regions indicate the area of the region.
Write the following in terms of A and B :

- a $\int_{-5}^1 f(x) dx$
- b $\int_{-5}^{-2} 3f(x) dx - \int_{-2}^1 f(x) dx$
- c $\left| \int_{-5}^1 f(x) dx \right|$
- d $\int_{-5}^1 |f(x)| dx$
- e $\int_{-5}^1 (f(x) + x) dx$ given that the definite integral $\int_{-5}^1 x dx = -12$.



- 35** Consider the function $f(x)$ as shown. The numbers in the shaded regions indicate the area of the region.
Write the following in terms of A , B and C :

- a $\int_0^6 f(x) dx$
- b $\int_6^{-1} f(x) dx$
- c The area bounded by the curve and the x -axis.
- d $\int_{-1}^6 |f(x)| dx$
- e $\int_{-1}^6 (2x - f(x)) dx$ given that the definite integral $\int_{-1}^6 2x dx = 35$



- 36** Consider the function $f(x)$ where $x = -4$ and 3 are the only x -intercepts and $\int_{-4}^1 f(x) dx = 4$ and $\int_1^3 f(x) dx = -7$.



Find the following:

- a** $\int_{-4}^3 f(x) dx$
- b** $\int_{-4}^3 |f(x)| dx$ of $f(x)$.
- c** The area bounded by the curve of $f(x)$ and the x -axis.
- d** $\int_3^{-4} (f(x) - x^3) dx$, given that $\int_3^{-4} x^3 dx = \frac{175}{4}$.

- 37** Consider the function $f(x)$ where $x = -2, 2$ and 8 are the only x -intercepts and $\int_{-2}^2 f(x) dx = -5$ and $\int_{-2}^8 f(x) dx = 3$.

Find the following:

- a** $\int_2^8 f(x) dx$
- b** The area bounded by the curve and the x -axis.
- c** $\int_{-2}^8 (2f(x) - 6x) dx$, given that $\int_{-2}^8 x dx = 30$.

- 38** Consider the function $f(x)$ where $x = -6, -2, 2$ and 7 are the only x -intercepts and $\int_{-6}^{-2} f(x) dx = -A$, $\int_{-2}^2 f(x) dx = B$ and $\int_2^7 f(x) dx = -C$.

Write the following in terms of A, B and C :

- a** $\left| \int_{-6}^7 f(x) dx \right|$
- b** The area bounded by the curve and the x -axis.
- c** $\int_{-2}^7 f(x) dx - \int_{-6}^2 f(x) dx$
- d** $\int_{-6}^2 2f(x) dx + \int_2^7 \frac{f(x)}{2} dx$
- e** $\int_{-6}^7 (3 - f(x) + x) dx$, given that $\int_{-6}^7 (3 + x) dx = \frac{91}{2}$.